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Algorithms and Probability **Peer Sheet P2** **FS 2026**

The solutions for this sheet must be submitted on Moodle by 26 March, 15:59. Afterwards you are assigned solutions of another student, which you have to grade (and upload to Moodle) until 11:59 of the following TUESDAY.

Exercise P2.1 – *Colorful April*

In order to break up the gray of everyday life, the physics department of ETH has decided to create the campaign "Make My April More Colorful". They have just emailed out a long list of potential activities to all of their students. They ask students to pick at most two activities from the list that they would like to do on one evening in April. Students have been promised that every activity that at least one student picks will be organized on some evening in April. Two activities may be scheduled on the same day, unless there is a student that has requested both activities.

Once the emails have been sent out, they start to worry that perhaps they promised too much. Will it really be possible to keep what they have promised?

(a) Describe how to model this as a graph problem. More precisely, given the answers from all students, describe how to construct a graph such that the scheduling above is possible if and only if your constructed graph has a certain (which?) property, and prove your assertion.

(b) Suppose we run the greedy coloring algorithm on a graph G, with an arbitrary vertex order. Prove that if the algorithm uses k colors, then for any unordered pair $\{a, b\}$ in $\binom{[k]}{2}$, there is an edge in G whose end-points have colors a and b.

(c) Show that any graph G with chromatic number $\chi(G)$ has at least $\binom{\chi(G)}{2}$ edges.

(d) Suppose the physics department has at most 450 students. Show that, no matter what answers the students give, it will always be possible to schedule all activities during April.